AAE 507: Principles of Dynamics
Problem Set 7 & 8

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Out: Oct 23, 2026
PS7 Due: Oct 30, 2026 at 11:59 PM ET
PS8 Due: Nov 6, 2026 at 11:59 PM ET
(submission at Brightspace>Content>Gradescope)

Use 1×10^{-12} for the integration tolerance of ordinary differential equations unless otherwise requested. When plotting trajectories in a 2-D or 3-D space, use equal axes for the x-, y-, and z-axes.

Problem Set 7 Hamiltonian Mechanics Foundations

(a): **[AI use Level: 0] Non-autonomous harmonic oscillator.**

Consider a one-dimensional oscillator with generalized coordinate q and a time-varying stiffness:

$$L(q, \dot{q}, t) = \frac{1}{2} m \dot{q}^2 - \frac{1}{2} k(t) q^2, \qquad k(t) = k_0 \left[1 + \epsilon \sin(\omega t)\right], \tag{1}$$

where $m, k_0, \omega > 0$ and $0 \leq \epsilon < 1$.

(a.1): **Calculate** the conjugate momentum p, and **invert** the result to express $\dot{q}$ in terms of p.

(a.2): Using a Legendre transformation, **construct** $H(q, p, t)$. Then **derive** Hamilton's canonical equations and **show** that they reproduce the Lagrange's equation obtained from Eq. (1).

(a.3): **Show** that $\frac{\mathrm{d}H}{\mathrm{d}t} = \frac{\partial H}{\partial t}$. **Discuss** whether H is conserved, and identify the condition under which it would be conserved.

(a.4): Introduce the extended phase-space variables

$$q_0 = t - t_0, \qquad \widehat{H}(q, p, q_0, p_0) = H(q, p, q_0) + p_0, \quad \text{with} \quad p_0(t_0) = -H(t_0) \tag{2}$$

Derive the Hamilton's canonical equations. **Show** that $p_0(t) = -H(t)$ and $\widehat{H}(t) = 0$ for all t.

(a.5): **Optional (extra credit)**: **Derive** the same Hamilton's equations from $\hat{H}$ using the variational approach, particularly the Hamilton's principle.

(b): **[AI use Level: 0] Earth-orbiting spacecraft in Cartesian Coordinates**

Consider a spacecraft of mass m with inertial Cartesian position: $\boldsymbol{r} = x\hat{\boldsymbol{e}}_x + y\hat{\boldsymbol{e}}_y + z\hat{\boldsymbol{e}}_z$ in the perturbed two-body environment due to J_2 and atmospheric drag. The potential energy is given by

$$V(\boldsymbol{r}) = -\frac{\mu m}{r} + \frac{\mu m J_2 R_E^2}{2r^5}\left(3z^2 - r^2\right). \tag{3}$$

Model atmospheric drag using the same approach as in PS1:

$$\boldsymbol{F}_D = -\frac{1}{2} C_D A \rho(r) \left\|\boldsymbol{v}\right\|_2 \boldsymbol{v}, \qquad \boldsymbol{v} \coloneqq \dot{\boldsymbol{r}}, \qquad \rho(r) = \rho_0 \exp\left[-\frac{r - r_{\mathrm{ref}}}{H_\rho}\right]. \tag{4}$$

All constants in Eqs. (3) and (4) are positive. Take $\boldsymbol{r}$ as the generalized coordinates $\boldsymbol{q}$.

(b.1): **Form** the Lagrangian $L(\boldsymbol{q}, \dot{\boldsymbol{q}})$, and **derive** the generalized momenta $\boldsymbol{p}$ as a function of $\boldsymbol{q}, \dot{\boldsymbol{q}}$.

(b.2): **Form** the Hamiltonian and **express** it as a function of $\boldsymbol{q}, \boldsymbol{p}$, i.e., $H(\boldsymbol{q}, \boldsymbol{p})$.

(b.3): Recall from class that the process for constructing Hamiltonian becomes largely simplified if the kinetic energy takes the form: $T(\boldsymbol{q}, \dot{\boldsymbol{q}}, t) = T_0(\boldsymbol{q}, t) + \boldsymbol{a}(\boldsymbol{q}, t)^\top \dot{\boldsymbol{q}} + \frac{1}{2}\dot{\boldsymbol{q}}^\top [\mathbf{I}(\boldsymbol{q}, t)]\dot{\boldsymbol{q}}$. **Show** that the system considered here meets this condition, **construct** the Hamiltonian exploiting this fact, and **verify** that the resulting Hamiltonian is equivalent to the one obtained in the previous question.

(b.4): **Derive** the equations of motion from H using the Hamilton's equations.

(b.5): **Derive** the rate of change of the Hamiltonian for this system. **Discuss** what drives the Hamiltonian change in this system, and how it differs from the harmonic oscillator example above.

(b.6): **Discuss** what quantities are conserved when drag is not present and why. Also, **discuss** why drag changes that conservation property.

Problem Set 8 Canonical Transformations

(a): **[AI use Level: 0] Symplectic condition.**

Consider a system whose dynamics are described by the following Hamiltonian:

$$H(q, p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2, \qquad \boldsymbol{x} = \begin{bmatrix} q \\ p \end{bmatrix} \tag{5}$$

(a.1): **Derive** the equations of motion using Hamilton's equations and **confirm** that it represents a harmonic oscillator.

(a.2): For some constant $a \neq 0$, consider the coordinate–momentum scaling

$$q^* = aq, \qquad p^* = \frac{p}{a}. \tag{6}$$

Write the inverse transformation $\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{y})$, where $\boldsymbol{y} = [q^*, p^*]^\top$. **Discuss** whether this transformation is canonical or not using the symplectic condition:

$$F^\top J F = J, \qquad F = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{y}}, \qquad J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \tag{7}$$

(a.3): Repeat the test for $q^* = aq$, $p^* = ap$. **Determine** the values of a for which this transformation is canonical (in the sense of the second condition covered in class).

(a.4): Use the type-2 generating function

$$W_2(q, p^*) = aqp^* \tag{8}$$

to **derive** Eq. (6). For $a = \sqrt{m\omega}$, **construct** the transformed Hamiltonian $K(q^*, p^*)$ and **derive** its canonical equations. **Discuss** an advantage of the transformed form.

(b): **[AI use Level: 1] Canonical Cartesian-to-spherical transformation.**

Let us consider a canonical transformation of the same space example as in PS7(b) to the spherical coordinates (r, ϕ, θ), where r, ϕ, θ are defined in class. Let the old canonical variables be $\boldsymbol{q} = [x, y, z]^\top$ and $\boldsymbol{p} = [p_x, p_y, p_z]^\top$. Define the new coordinates and corresponding momenta as follows:

$$\boldsymbol{q}^* = [r, \phi, \theta]^\top, \quad \boldsymbol{p}^* = [p_r^*, p_\phi^*, p_\theta^*]^\top \tag{9}$$

(b.1): **Express** $\boldsymbol{q}$ as a function of $\boldsymbol{q}^*$ explicitly, and write it as $\boldsymbol{q} = \boldsymbol{f}(\boldsymbol{q}^*)$.

(b.2): **Derive** the new momenta $\boldsymbol{p}^*$ using the type-3 generating function:

$$W_3(\boldsymbol{q}^*, \boldsymbol{p}) = -\boldsymbol{p}^\top \boldsymbol{f}(\boldsymbol{q}^*) \tag{10}$$

(b.3): **Show** that

$$\det\left(\frac{\partial^2 W_3}{\partial \boldsymbol{q}^* \partial \boldsymbol{p}}\right) = -r^2 \cos\theta. \tag{11}$$

Identify the configurations at which this generating-function construction is not locally invertible, and **explain** the geometric reason.

(b.4): Out of the other three common types of generating functions covered in class, **discuss** if any other type(s) can provide the same $\boldsymbol{p}^*$, and if so, **derive** $\boldsymbol{p}^*$ using the other generating function(s).

(b.5): Starting from the Cartesian Hamiltonian in PS7(b), **construct** its transformed Hamiltonian $K(\boldsymbol{q}^*, \boldsymbol{p}^*)$.

(b.6): **Derive** the equations of motion in $(\boldsymbol{q}^*, \boldsymbol{p}^*)$ using Hamilton's equations, where be sure to transform the external force (drag) appropriately under this canonical transformation. **Discuss** whether the drag term appears in the equations for $\dot{\boldsymbol{q}}^*$ and why.

(b.7): **Identify** the cyclic coordinate and **show** that its conjugate momentum is the z-component of the inertial angular momentum.

(b.8): **Optional (extra credit)**:

Let us verify we obtain the same spherical-coordinate Hamiltonian by constructing it from scratch. **Form** the Lagrangian in the spherical coordinates, **calculate** the conjugate momenta, and **perform** the Legendre transformation. **Verify** that this gives the same $K(\boldsymbol{q}^*, \boldsymbol{p}^*)$ as the canonical transformation.